Tashkent Institute of Textile and Light Industry
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Editor-in-Chief
IEEE Internet of Things Journal

Dear Editor-in-Chief,

We submit the manuscript *"An IoT-Enabled Hybrid Soft Sensor for Real-Time Residual-Moisture Monitoring and Resource Accounting in Compact Wool Scouring Machines"* for consideration as a **Regular Paper** in the IEEE Internet of Things Journal. We request **Traditional (non-open-access) publication**.

The paper addresses IoT trials/experiments and IoT application design, both within the scope of the journal. Its main contributions are: (i) a low-cost ESP32-based IoT architecture for wool scouring machines, which today operate without instrumentation; (ii) a hybrid soft sensor that uses a design-of-experiments model of the squeezing device as a physics-informed prior and is recalibrated online from sparse laboratory samples; (iii) real-time per-kilogram energy and water indicators and fault alarms; and (iv) a reproducible reanalysis of the historical mechanical-process baseline and an open-source implementation (https://github.com/nasin0764-bit/woolwash-iot) with a defined prospective field-validation protocol. [TODO: Update (iv) with the completed field validation before submission.]

The manuscript is original, has not been published previously, and is not under consideration elsewhere. [TODO: If any part was presented at a conference, cite it here and explain the extension.] All authors (R. Tuguzbayeva, S. Khakimov, F. Dustova, R. Rosulov, and M. Anorkulov) have approved the submission and declare no conflicts of interest.

Suggested reviewers (no conflict of interest, outside our institution):

1. [TODO: Name, affiliation, e-mail – industrial IoT / soft sensors]
2. [TODO: Name, affiliation, e-mail – textile machinery / wool processing]
3. [TODO: Name, affiliation, e-mail – edge computing for manufacturing]

Sincerely,

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on behalf of all co-authors
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